

PRATHAM P PANSARE

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ABOUT ME

Aspiring software developer with hands-on experience building scalable, full-stack applications and AI-driven platforms. Proficient in modern web frameworks, relational databases, and machine learning integration, with a strong ability to translate complex technical requirements into user-centric solutions. Passionate about software architecture, clean code, and continuous learning.

EDUCATION

B.Tech in Information Technology	A.P. Shah Institute of Technology, Thane
Higher Secondary Education	Jana Gana Mana College, Dombivli
Secondary Education	Model English School, Dombivli West

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, CSS, HTML
Frameworks:	React, Next.js, FastAPI, Flask, Tailwind CSS, Bootstrap
ML / AI:	TensorFlow, Google Colab
Databases:	PostgreSQL, MySQL, SQLite, Supabase
Cloud / DevOps:	AWS, Cloudflare R2, GitHub Actions, Vercel, HF Spaces
Tools:	Figma, Canva, GitHub

PROJECTS

ShetVaidya — AI Crop Disease Analyzer & Medicine Verifier

React, FastAPI, TensorFlow, PostgreSQL, Cloudflare R2

ShetVaidya ("field doctor" in Marathi) is an AI-powered platform designed to help farmers instantly diagnose crop diseases and verify medicine authenticity. The goal was to bridge the gap between rural agriculture and modern technology by providing real-time, location-aware diagnostic tools that reduce crop loss and prevent counterfeit medicine usage.

- Built a production-grade, location-aware crop disease analyzer and medicine verification platform to empower farmers with immediate diagnostic tools.
- Developed a TensorFlow MobileNetV2 inference microservice with prediction safety — dedicated non-leaf/garbage class rejection and confidence gating for out-of-scope images.
- Engineered a microservice architecture (frontend, backend, ML service) with FastAPI, JWT HttpOnly cookies, Google OAuth, and API rate limiting on auth and batch verification flows.
- Configured Cloudflare R2 object storage for secure image uploads (5MB limit, MIME validation) and integrated barcode scanning for medicine batch verification with non-public batch codes.
- Established CI/CD via GitHub Actions for containerized deployment across Vercel (frontend) and Hugging Face Spaces (backend + ML), with PostgreSQL for data persistence.

Khatawali Records — Credit/Debit Ledger App

React, Supabase, Bootstrap

Khatawali Records is a mobile-first ledger application built for small business owners and individuals who struggle with tracking credit and debit transactions. The app simplifies financial record-keeping by enabling category-separated tracking, quick contact imports, and secure per-user data isolation — all packaged as a native Android app.

- Engineered a mobile-first credit/debit ledger application for small business owners and individuals, with category-separated tracking for the same person across independent ledgers.
- Packaged natively for Android with deep-link OAuth callbacks, in-app contact importer, and biometric fingerprint lock configurable via app settings.
- Designed a PostgreSQL schema on Supabase with Row Level Security (RLS) policies and Supabase Auth (Google OAuth) to guarantee per-user data isolation and privacy.
- Built a dashboard with advanced search (name, phone, description, category), date-range and year/month filtering, and quick actions for call, WhatsApp, receive, give, and transfer.
- Integrated per-person PDF and Excel export using autotable and PapaParse, along with profile/business details, bank holidays, recycle bin, and developer support.

Scholar Nexus — AI Research Discovery Platform

Next.js, TypeScript, Prisma, D3.js

Scholar Nexus addresses the challenge of fragmented academic research by aggregating papers from 9 major databases into a single intelligent platform. It helps researchers discover relevant literature faster through AI-driven query expansion, smart deduplication, and visual citation networks — turning hours of manual searching into minutes of guided exploration.

- Architected an AI-powered research paper discovery platform that simultaneously queries 9 academic databases (Semantic Scholar, arXiv, Crossref, PubMed, OpenAlex, and more) in parallel.
- Implemented a 10-step search pipeline: query understanding, AI-driven expansion, parallel fanout, normalization, deduplication (DOI + title-hash + fuzzy), ranking, and AI insights.
- Developed composite ranking scoring papers across citation impact, recency, source authority, and open-access availability, with AI-generated per-paper insights (TL;DR, methodology, limitations).
- Built interactive D3.js citation network graphs, PDF full-text Q&A; chatbot, evidence synthesis across papers, color-coded reading collections, daily/weekly search alerts, and author profiles.
- Engineered 18 API route handlers with rate limiting (token bucket), SSRF protection, Zod input validation, and Prisma parameterized queries; deployed on Vercel with GitHub Actions.

Air Pollution Tracker — Environmental Intelligence

React, Tailwind CSS, Supabase, React Leaflet, Recharts

This project tackles the lack of accessible, real-time air quality data for citizens and government bodies alike. By visualizing live AQI metrics for 200+ cities with role-based portals, it empowers citizens to make health-informed decisions while giving officials the analytical tools needed for environmental policy and incident response.

- Created a data-driven React application visualizing live and historical air-quality metrics for 200+ cities worldwide, with OpenAQ v3 integration and EPA breakpoint AQI computation.
- Implemented differentiated role-based portals: citizen experience (dashboard, AQI tracking, map view) and government portal (heatmaps, incident desk, pollutant intelligence, reporting panels).
- Built interactive map views with React Leaflet clustered markers and geolocation (browser API with IP-based fallback via ipwho.is), plus Phase 5 analytics for single-city deep dives and forecasting.
- Designed a robust data layer using React Query v5 with stale-while-revalidate caching, 30-minute localStorage cache for OpenAQ snapshots, and virtualized list rendering via react-window.
- Integrated Supabase Auth with domain-guardrailed registration (citizen: @gmail/@outlook, government: @gov), email OTP verification, offline-friendly preferences, and audit logging for government.